

Final-Solutions

ECON301
Spring 2021



Question 1

(a) Since $E(Y_i) = \theta$ and $\text{Var}(Y_i) = \theta(1-\theta)$ and we have a random sample, the conditions of the WLLN (Thm 4.4) apply so we have that $\hat{\theta}_n = \frac{1}{n} \sum_{i=1}^n Y_i \rightarrow_p E(Y_i) = \theta$.

(b) Since $E(Y_i) = \theta$, $\text{Var}(Y_i) = \theta(1-\theta)$ and we have a random sample, the CLT (Thm 4.5) implies that $\frac{\hat{\theta}_n - \theta}{\sqrt{\theta(1-\theta)/n}} \rightarrow_d N(0,1)$, which is equivalent to $\sqrt{n}(\hat{\theta}_n - \theta) \rightarrow_d N(0, \theta(1-\theta))$.

(c) Let $\hat{\gamma}_n = \hat{\theta}_n / (1 - \hat{\theta}_n)$, and define $g(x) = \frac{x}{1-x}$; g is continuous for all $x \in (0,1)$, and since $\theta \in (0,1)$ by hypothesis, we can use Thm. 4.3 to get: $\hat{\gamma}_n = g(\hat{\theta}_n) = \frac{\hat{\theta}_n}{1 - \hat{\theta}_n} \rightarrow_p g(\theta) = \frac{\theta}{1-\theta} = \gamma$.

(d) We know from (b) that: $\sqrt{n}(\hat{\theta}_n - \theta) \rightarrow_d N(0, \theta(1-\theta))$ and $\hat{\gamma}_n = g(\hat{\theta}_n)$, since g is continuously differentiable on the interval $(0,1)$ and $g'(x) = \frac{0(1-x) - x(-1)}{(1-x)^2} = \frac{1-x+x}{(1-x)^2} = \frac{1}{(1-x)^2}$, we can use the Theorem as stated to get

$$\sqrt{n}(\hat{\gamma}_n - \gamma) \rightarrow_d N(0, \theta(1-\theta) \frac{1}{(1-\theta)^4}) = N(0, \frac{\theta}{(1-\theta)^3})$$

(e) Let $s_n^2 = \frac{\hat{\theta}_n}{(1-\hat{\theta}_n)^3}$, since $\hat{\theta}_n \rightarrow_p \theta$, Theorem 4.3 applies: define $h(x) = \frac{x}{(1-x)^3}$, h is continuous on $x \in (0,1)$ so we have that $s_n^2 = h(\hat{\theta}_n) \rightarrow_p h(\theta) = \frac{\theta}{(1-\theta)^3}$.

(f) Define $Z_n = \frac{\hat{\gamma}_n - \gamma}{\sqrt{\frac{\theta}{n(1-\theta)^3}}}$, by (d) $Z_n \rightarrow_d N(0,1)$. Applying Thm. 4.3 and (e) yields $\sqrt{s_n^2} \rightarrow_p \sqrt{\frac{\theta}{(1-\theta)^3}}$.

Define $W_n = \frac{\sqrt{s_n^2}}{\sqrt{\frac{\theta}{(1-\theta)^3}}} \rightarrow_p \frac{\sqrt{\frac{\theta}{(1-\theta)^3}}}{\sqrt{\frac{\theta}{(1-\theta)^3}}} = 1 \neq 0$ which was another application of Thm 4.3. Note: we write $s_n = \sqrt{s_n^2}$

Define: $T_n = \frac{Z_n}{W_n} = \frac{\frac{\hat{\theta}_n - \theta}{\sqrt{\frac{\theta}{n(1-\theta)^3}}}}{\frac{\sqrt{s_n^2}}{\sqrt{\frac{\theta}{(1-\theta)^3}}}} = \frac{\hat{\theta}_n - \theta}{\sqrt{\frac{s_n^2}{n}}} = \frac{\hat{\theta}_n - \theta}{\frac{s_n}{\sqrt{n}}}$, by Slutsky's Thm (Thm 4.6) part (iii) $T_n \xrightarrow{d} \frac{N(0,1)}{1} = N(0,1)$

T_n is the asymptotic pivot because all of it can be estimated.

(g) Let $z_{\alpha/2}$ be the $(1-\alpha)\%$ critical value for the CI, then

$$P(|T_n| \leq z_{\alpha/2}) \approx 1 - \alpha \Leftrightarrow P(\hat{\theta}_n - z_{\alpha/2} \cdot s_n \leq \theta \leq \hat{\theta}_n \leq \hat{\theta}_n + z_{\alpha/2} s_n) \approx 1 - \alpha$$

Question 2 (There are several ways to solve this, here I'm presenting a solution using matrix algebra)

The regressor matrix for model (1) is $X = [\underline{1} \ \underline{x}_1 \ \underline{x}_2]$, whereas the regression matrix for model (2) is $X_* = [\underline{1} \ \underline{x}_1^* \ \underline{x}_2]$. Then:

$$\hat{\underline{\beta}} = (X^T X)^{-1} X^T \underline{y} \quad \text{and} \quad \hat{\underline{\delta}} = (X_*^T X_*)^{-1} X_*^T \underline{y}$$

Now, define the 3×3 matrix A as follows: $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c & 0 \\ 0 & 0 & 1 \end{bmatrix}$. Note that A is invertible, i.e., A^{-1} exists.

Then, we have that:

$$X A = [\underline{1} \ \underline{x}_1 \ \underline{x}_2] \begin{bmatrix} 1 & 0 & 0 \\ 0 & c & 0 \\ 0 & 0 & 1 \end{bmatrix} = [\underline{1} \ c \underline{x}_1 \ \underline{x}_2] = [\underline{1} \ \underline{x}_1^* \ \underline{x}_2] = X_* \quad \text{since} \quad \underline{x}_1^* = c \underline{x}_1$$

$$\begin{aligned} \text{Then} \quad \hat{\underline{\delta}} &= (X_*^T X_*)^{-1} X_*^T \underline{y} = [(XA)^T XA]^{-1} (XA)^T \underline{y} \\ &= [A^T X^T X A]^{-1} A^T X^T \underline{y} \quad \text{using properties of the transpose } ^T \text{ (Woodbridge D.2d)} \\ &= [(X^T X A)^{-1} (A^T)^{-1}] A^T X^T \underline{y} \\ &= A^{-1} (X^T X)^{-1} (A^T)^{-1} A^T X^T \underline{y} \quad \left. \begin{aligned} &\text{using properties of the inverse matrix (Woodbridge D.2g)} \\ &\text{using properties of the transpose } ^T \end{aligned} \right\} \\ &= A^{-1} (X^T X)^{-1} X^T \underline{y} \\ &= A^{-1} \hat{\underline{\beta}} \quad \text{since} \quad \hat{\underline{\beta}} = (X^T X)^{-1} X^T \underline{y} \end{aligned}$$

$$\text{Since} \quad A^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/c & 0 \\ 0 & 0 & 1 \end{bmatrix} \Rightarrow \quad \hat{\underline{\delta}} = \begin{pmatrix} \hat{\delta}_0 \\ \hat{\delta}_1 \\ \hat{\delta}_2 \end{pmatrix} = \begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1/c \\ \hat{\beta}_2 \end{pmatrix} \quad \therefore \quad \hat{\delta}_0 = \hat{\beta}_0, \quad \hat{\delta}_1 = \hat{\beta}_1/c, \quad \hat{\delta}_2 = \hat{\beta}_2$$

Question 3

(a) We are testing $H_0: \beta_2 = 0$ vs $H_1: \beta_2 \neq 0$. The test statistic is $t_{calc} = \frac{-2.195}{0.524} = -4.19$

The p-value is $P(|N(0,1)| > | -4.19 |) = 0.00027$.

Clearly the null is rejected at the 10%, 5% and 1% significance levels. (Note, you could also have read this immediately from the given table by looking at the # of *.)

$\therefore$ There is evidence that $hsize^2$ should be included.

The optimal high-school size is: $HS^o = -\frac{19.297}{2(-2.195)} = 4.3957$, i.e., around 439-440 students.

Here I used the formula in slide 6 of Lecture 7.

(b) We are looking for the following difference:

$$E^*(sat | hsize = H, female = 1, black = 0) = \beta_0 + \beta_1 H + \beta_2 H^2 + \beta_3 \quad \therefore \text{The estimated difference is } \hat{\beta}_3 = -45.091$$

$$E^*(sat | hsize = H, female = 0, black = 0) = \beta_0 + \beta_1 H + \beta_2 H^2$$

The difference is $= \beta_3$

To test the proposed hypothesis: $H_0: \beta_3 = 0$ vs $H_1: \beta_3 \neq 0$. The test statistic is $t_{calc} = \frac{-45.091}{4.226} = -10.66$

The critical value for this test is $Z_{0.01/2} = 2.575829$ and the rejection region is $R = \{|t_{calc}| > 2.575829\}$

Clearly, the null is rejected at the 1% level. (Note, this information was also readily available from the table.)

(c) We are looking for the following difference:

$$E^*(\text{sat} | \text{hsize} = H, \text{female} = 0, \text{black} = 0) = \beta_0 + \beta_1 H + \beta_2 H^2$$

$$E^*(\text{sat} | \text{hsize} = H, \text{female} = 0, \text{black} = 1) = \beta_0 + \beta_1 H + \beta_2 H^2 + \beta_4$$

The Difference is $= \beta_4$

The hypothesis to test is:

$H_0: \beta_4 = 0$ vs $H_1: \beta_4 \neq 0$. The test statistic is

$$t_{\text{calc}} = \frac{-169.813}{15.423} = -11.01$$

The critical value for this test is $z_{0.05/2} = 1.959964$

and the rejection region is $R = \{|t_{\text{calc}}| > 1.959964\}$.

$\therefore$ We reject the null at the 5% level.

(d) We are looking for the following difference:

$$E^*(\text{sat} | \text{hsize} = H, \text{female} = 1, \text{black} = 1) = \beta_0 + \beta_1 H + \beta_2 H^2 + \beta_3 + \beta_4 + \beta_5$$

$$E^*(\text{sat} | \text{hsize} = H, \text{female} = 1, \text{black} = 0) = \beta_0 + \beta_1 H + \beta_2 H^2 + \beta_3$$

The difference is $= \beta_4 + \beta_5$

The hypothesis to test is:

$H_0: \beta_4 + \beta_5 = 0$ vs $H_1: \beta_4 + \beta_5 \neq 0$. The test statistic is

$$t_{\text{calc}} = \frac{\hat{\beta}_4 + \hat{\beta}_5}{SE(\hat{\beta}_4 + \hat{\beta}_5)}$$

There are several ways to compute the denominator. I present one here:

$$SE(\hat{\beta}_4 + \hat{\beta}_5) = (\widehat{Var}(\hat{\beta}_4) + \widehat{Var}(\hat{\beta}_5) + 2\widehat{Cov}(\hat{\beta}_4, \hat{\beta}_5))^{1/2} = (237.863505 + 372.702161 - 2(237.367652))^{1/2} = 11.65463$$

$\therefore t_{\text{calc}} = (-169.913 + 62.306) / 11.65463 = -9.224343$. We know that the rejection region for this

two-sided test is $R = \{|t_{\text{calc}}| > 2.575829\}$ so we clearly reject the null at the 1% level.

(e) We are looking for the following difference:

We estimate Δsat by $\Delta \hat{\text{sat}} = \hat{\beta}_1 + 5\hat{\beta}_2 - \hat{\beta}_4 - \hat{\beta}_5$.

$$\text{sat}_A = E^*(\text{sat} | \text{hsize} = 2, \text{female} = 1, \text{black} = 1) = \beta_0 + 2\beta_1 + 4\beta_2 + \beta_3 + \beta_4 + \beta_5$$

We also need $SE(\Delta \hat{\text{sat}}) = SE(\hat{\beta}_1 + 5\hat{\beta}_2 - \hat{\beta}_4 - \hat{\beta}_5)$. Define $\Delta^T = (0, 1, 5, 0, -1, -1)$. Then $\Delta \hat{\text{sat}} = 115.829$ and

$$\text{sat}_B = E^*(\text{sat} | \text{hsize} = 3, \text{female} = 1, \text{black} = 0) = \beta_0 + 3\beta_1 + 9\beta_2 + \beta_3$$

$SE(\Delta \hat{\text{sat}}) = \sqrt{\Delta^T \hat{\Omega} \Delta} = 11.66355$. Then 95% is $[\Delta \hat{\text{sat}} \pm z_{0.05/2} SE(\Delta \hat{\text{sat}})] = [92.96907, 138.6893]$

$$\Delta \text{sat} = \text{sat}_B - \text{sat}_A = \beta_1 + 5\beta_2 - \beta_4 - \beta_5$$

Question 4

- (a) We test two hypotheses: i) $H_0: \beta_0 = 0$ vs $H_1: \beta_0 \neq 0$
 ii) $H_0: \beta_1 = 1$ vs $H_1: \beta_1 \neq 1$

The respective test statistics are: $t_{\text{calc}, i} = \frac{-14.472}{20.382} = -0.710$

and $t_{\text{calc}, ii} = \frac{0.976 - 1}{0.067} = -0.3662$. The respective p-values

are $P_i = \Pr(|N(0,1)| > | -0.710 |) = 0.4777$

$P_{ii} = \Pr(|N(0,1)| > | -0.3662 |) = 0.72019$

Thus, we don't reject the null in both cases i) AND ii) at any significance level.

Although we may be tempted to conclude that the assessment is rational based on the results above, we CANNOT do this, using the two disjoint tests we have just carried out: rationality should be tested as a joint statement, $\beta_0 = 0$ AND $\beta_1 = 1$

- (b) Define $H = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ and $\Theta_{HIP} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, then we will test

$H_0: H\beta = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ vs $H_1: H\beta \neq \begin{pmatrix} 0 \\ 1 \end{pmatrix}$. The test

statistic is $W_{\text{calc}} = (H\hat{\beta} - \Theta_{HIP})^T (H\hat{\Omega}H^T)^{-1} (H\hat{\beta} - \Theta_{HIP})$

which yields $W_{\text{calc}} = 24.1287$. The p-value is

$\Pr(\chi^2(2) > 24.1287) = 0.0000058$. Therefore we reject the null at all relevant significance levels.

$\therefore$ We conclude that there is strong statistical evidence against the rationality of the assessment.

(c) See the comments in red.

(d) Define

$$H_1 = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \text{ and } \Theta_{HIP} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \text{ then the}$$

hypothesis to test is $H_0: H_1\beta = \Theta_{HIP}$ vs $H_1: H_1\beta \neq \Theta_{HIP}$

We again use the Wald statistic:

$$W_{\text{calc}} = (H_1\hat{\beta} - \Theta_{HIP})^T (H_1\hat{\Omega}H_1^T)^{-1} (H_1\hat{\beta} - \Theta_{HIP})$$

$$= (\hat{\beta}_2, \hat{\beta}_3, \hat{\beta}_4) \left([\hat{\Omega}(i,j)]_{\substack{i=3, \dots, 5 \\ j=3, \dots, 5}} \right)^{-1} \begin{pmatrix} \hat{\beta}_2 \\ \hat{\beta}_3 \\ \hat{\beta}_4 \end{pmatrix}$$

where $[\hat{\Omega}(i,j)]_{\substack{i=3, \dots, 5 \\ j=3, \dots, 5}}$ refers to the submatrix of $\hat{\Omega}$ starting at row and column 3 down. The calculation yields:

$W_{\text{calc}} = 18.13876$. The critical value for the χ^2 distribution with 3 degrees of freedom at the 5% level is $\chi_{0.05}^2(3) = 4.6448$. Thus, we reject the null at the 5% level, since $W_{\text{calc}} > 4.6448$